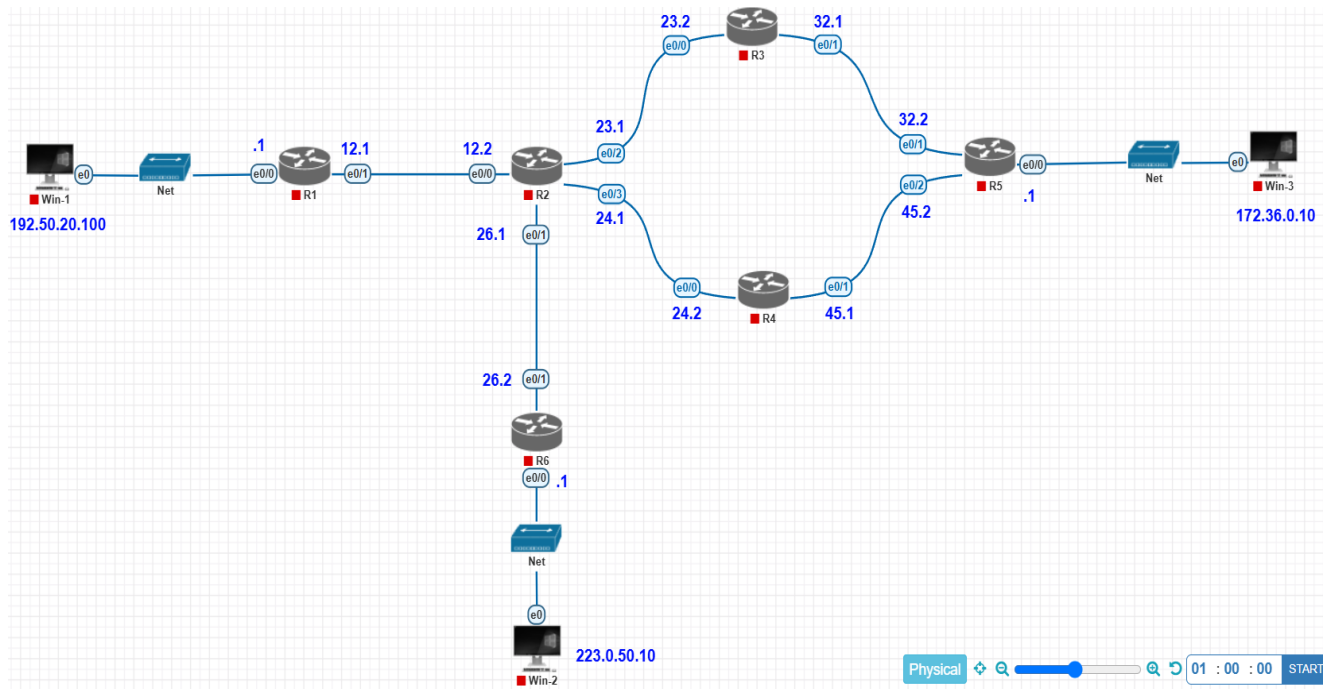


Static Routing Project 2



Router Configuration & Static Routing

```
R6#sh ip int brief
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 223.0.50.1 YES manual up
Ethernet0/1 26.0.0.2 YES manual up
Ethernet0/2 unassigned YES TFTP down
Ethernet0/3 unassigned YES TFTP down

R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
        s - application route
        * - replicated route, % - next hop override, p - overrides from PFH

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 26.0.0.1
S 23.0.0.0/8 [1/0] via 26.0.0.1
S 24.0.0.0/8 [1/0] via 26.0.0.1
C 26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 26.0.0.0/8 is directly connected, Ethernet0/1
L 26.0.0.2/32 is directly connected, Ethernet0/1
S 32.0.0.0/8 [1/0] via 26.0.0.1
S 45.0.0.0/8 [1/0] via 26.0.0.1
S 172.36.0.0/16 [1/0] via 26.0.0.1
L 192.50.20.0/24 [1/0] via 26.0.0.1
C 223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C 223.0.50.0/24 is directly connected, Ethernet0/0
L 223.0.50.1/32 is directly connected, Ethernet0/0
R6#
```

```
R5#sh ip int brief
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 172.36.0.1 YES manual up
Ethernet0/1 32.0.0.2 YES manual up
Ethernet0/2 45.0.0.2 YES manual up
Ethernet0/3 unassigned YES TFTP down

R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
        s - application route
        * - replicated route, % - next hop override, p - overrides from PFH

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 45.0.0.1
S 23.0.0.0/8 [1/0] via 45.0.0.1
S 24.0.0.0/8 [1/0] via 45.0.0.1
S 26.0.0.0/8 [1/0] via 45.0.0.1
C 32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 32.0.0.0/8 is directly connected, Ethernet0/1
L 32.0.0.2/32 is directly connected, Ethernet0/1
C 45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 45.0.0.0/8 is directly connected, Ethernet0/2
L 45.0.0.2/32 is directly connected, Ethernet0/2
S 172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.36.0.0/16 is directly connected, Ethernet0/0
L 172.36.0.1/32 is directly connected, Ethernet0/0
S 192.50.20.0/24 [1/0] via 45.0.0.1
S 223.0.50.0/24 [1/0] via 45.0.0.1
R5#
```

```
R4(config)#
R4#
R4#sh ip int brief
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 24.0.0.2 YES manual up
Ethernet0/1 45.0.0.1 YES manual up
Ethernet0/2 unassigned YES TFTP down
Ethernet0/3 unassigned YES TFTP down

R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
        s - application route
        * - replicated route, % - next hop override, p - overrides from PFH

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 24.0.0.1
S 23.0.0.0/8 [1/0] via 24.0.0.1
C 24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 24.0.0.0/8 is directly connected, Ethernet0/0
L 24.0.0.2/32 is directly connected, Ethernet0/0
S 26.0.0.0/8 [1/0] via 24.0.0.1
S 32.0.0.0/8 [1/0] via 24.0.0.1
C 45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 45.0.0.0/8 is directly connected, Ethernet0/1
L 45.0.0.1/32 is directly connected, Ethernet0/1
S 172.36.0.0/16 [1/0] via 45.0.0.2
S 192.50.20.0/24 [1/0] via 24.0.0.1
S 223.0.50.0/24 [1/0] via 24.0.0.1
R4#
```

```
R3
R3(config)#
*Jul 19 19:12:47.790: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int brief
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 23.0.0.2 YES manual up
Ethernet0/1 32.0.0.1 YES manual up
Ethernet0/2 unassigned YES TFTP down
Ethernet0/3 unassigned YES TFTP down
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
        a - application route
        * - replicated route, % - next hop override, p - overrides from PIR

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 32.0.0.2
C 23.0.0.0/8 is directly connected, Ethernet0/0
L 23.0.0.2/32 is directly connected, Ethernet0/0
S 24.0.0.0/8 [1/0] via 23.0.0.1
S 26.0.0.0/8 [1/0] via 23.0.0.1
S 32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 32.0.0.0/8 is directly connected, Ethernet0/1
L 32.0.1/32 is directly connected, Ethernet0/1
S 45.0.0.0/8 [1/0] via 23.0.0.1
S 172.36.0.0/16 [1/0] via 32.0.0.2
S 192.50.20.0/24 [1/0] via 23.0.0.1
S 223.0.50.0/24 [1/0] via 23.0.0.1
R3#
R3#
```

```
R2
R2#sh ip int brief
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 12.0.0.2 YES manual up
Ethernet0/1 26.0.0.1 YES manual up
Ethernet0/2 24.0.0.1 YES manual up
Ethernet0/3 24.0.0.1 YES manual up
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
        a - application route
        * - replicated route, % - next hop override, p - overrides from PIR

Gateway of last resort is not set

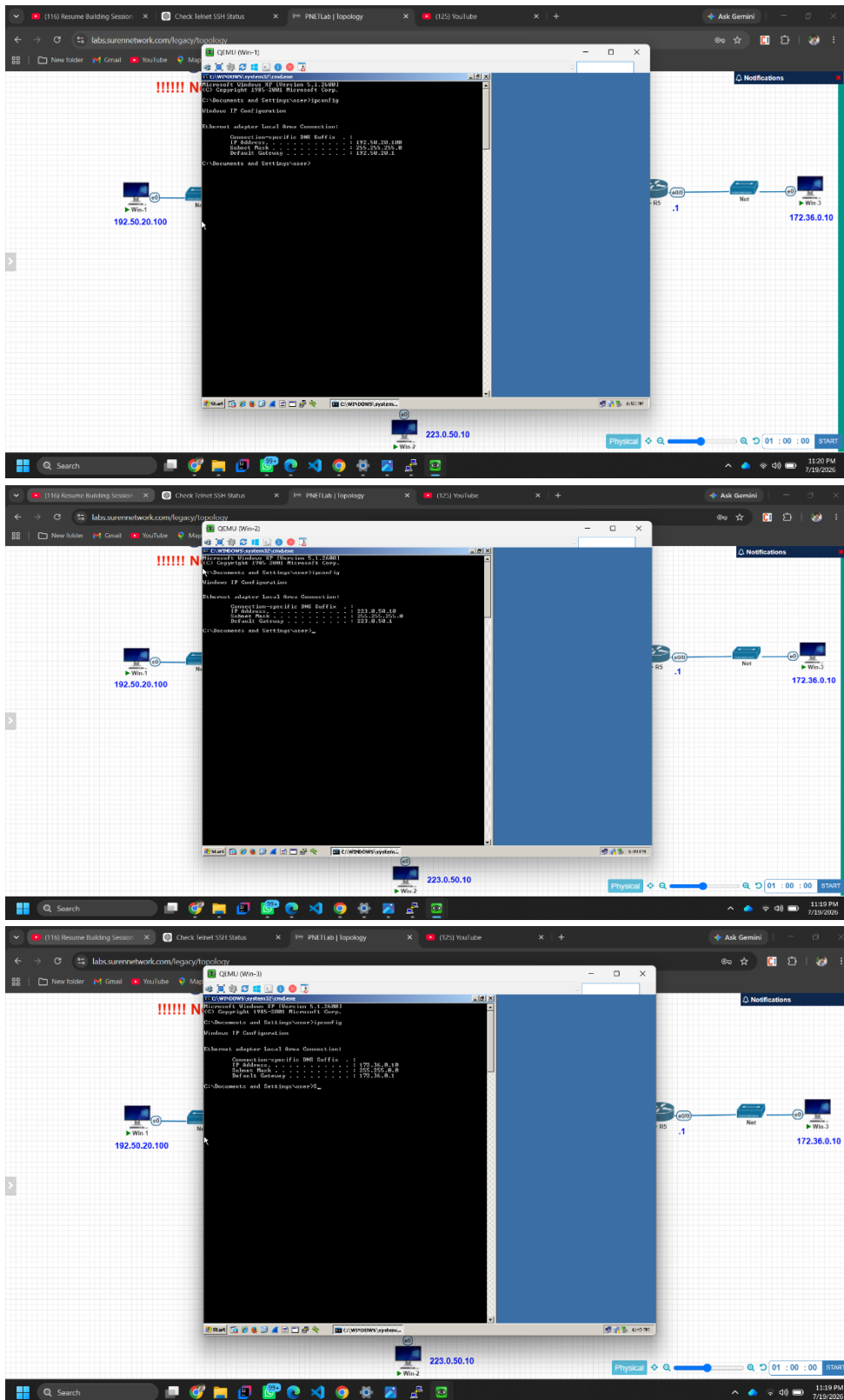
C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 12.0.0.0/8 is directly connected, Ethernet0/0
L 12.0.0.2/32 is directly connected, Ethernet0/0
C 23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 23.0.0.0/8 is directly connected, Ethernet0/2
L 23.0.1/32 is directly connected, Ethernet0/2
C 24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 24.0.0.0/8 is directly connected, Ethernet0/3
L 24.0.1/32 is directly connected, Ethernet0/3
C 26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 26.0.0.0/8 is directly connected, Ethernet0/1
L 26.0.1/32 is directly connected, Ethernet0/1
S 32.0.0.0/8 [1/0] via 23.0.0.2
S 45.0.0.0/8 [1/0] via 24.0.0.2
S 172.36.0.0/16 [1/0] via 24.0.0.2
S 192.50.20.0/24 [1/0] via 23.0.0.1
S 223.0.50.0/24 [1/0] via 26.0.0.2
R2#
R2#
```

```
R1
R1#sh ip int brief
Interface IP-Address OK? Method Status Prot
Loopback0 192.50.20.1 YES manual up
Ethernet0/0 12.0.0.1 YES manual up
Ethernet0/1 unassigned YES TFTP down
Ethernet0/2 unassigned YES TFTP down
Ethernet0/3 unassigned YES TFTP down
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
        a - application route
        * - replicated route, % - next hop override, p - overrides from PIR

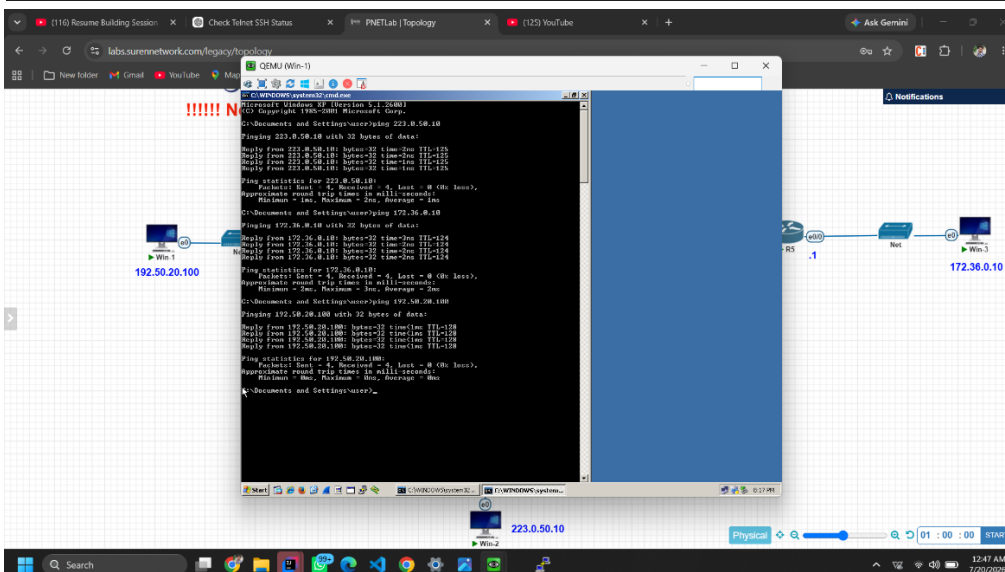
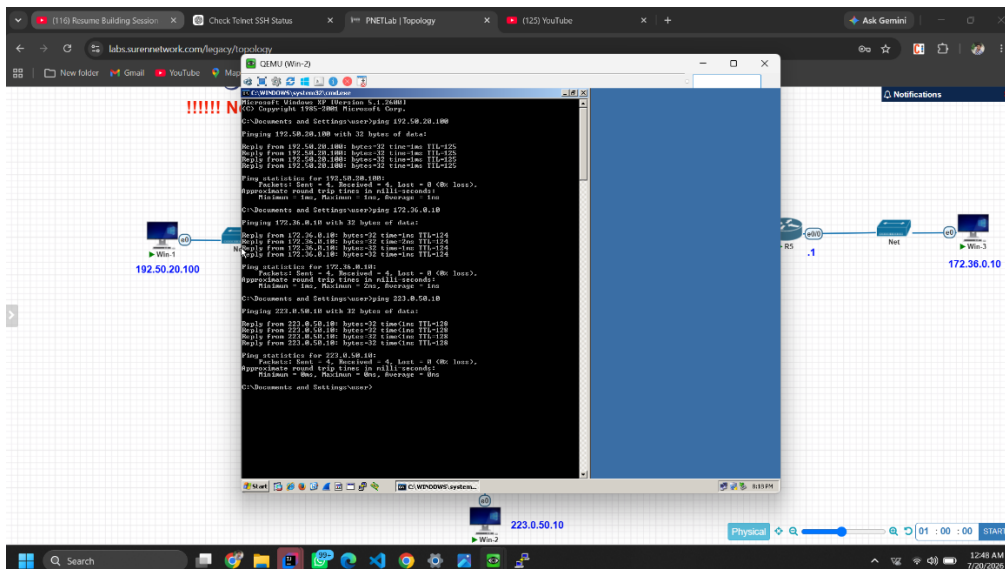
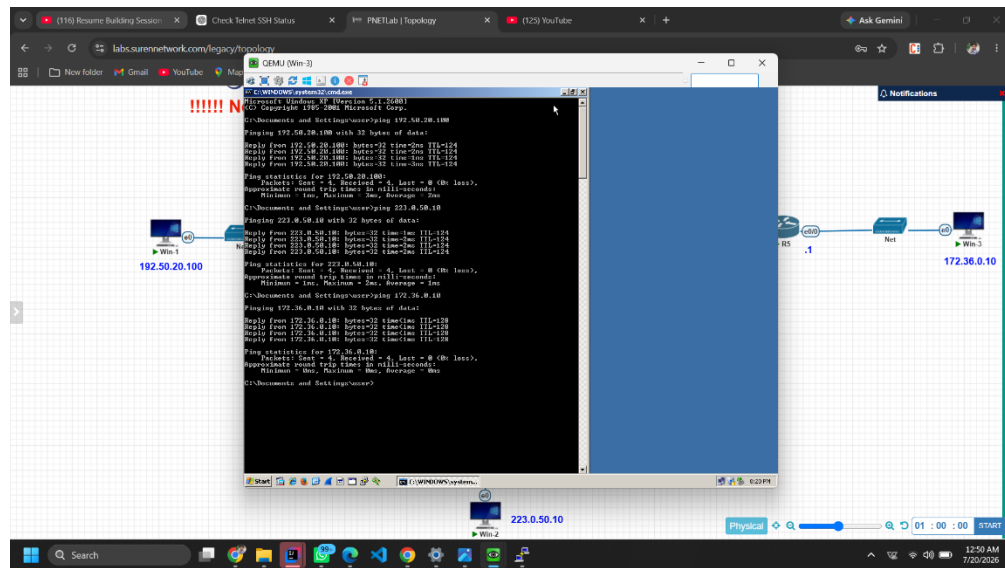
Gateway of last resort is not set

C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 12.0.0.0/8 is directly connected, Ethernet0/1
L 12.0.0.1/32 is directly connected, Ethernet0/1
S 23.0.0.0/8 [1/0] via 12.0.0.2
S 24.0.0.0/8 [1/0] via 12.0.0.2
S 26.0.0.0/8 [1/0] via 12.0.0.2
S 32.0.0.0/8 [1/0] via 12.0.0.2
S 45.0.0.0/8 [1/0] via 12.0.0.2
S 172.36.0.0/16 [1/0] via 12.0.0.2
S 192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.50.20.0/24 is directly connected, Ethernet0/0
L 192.50.20.1/32 is directly connected, Ethernet0/0
S 223.0.50.0/24 [1/0] via 12.0.0.2
R1#
R1#
```

Assign IP to the PC



Ping From PC



PING via source Interface

```
R5
R5#ping 12.0.0.1 source 45.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1, timeout is 2 seconds:
Packet sent with a source address of 45.0.0.2
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R5#ping 12.0.0.1 source 32.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1, timeout is 2 seconds:
Packet sent with a source address of 32.0.0.2
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R5#ping 12.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R5#
```

```
R2
R2#ping 172.36.0.1 source 23.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.36.0.1, timeout is 2 seconds:
Packet sent with a source address of 23.0.0.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/5 ms
R2#ping 172.36.0.1 source 24.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.36.0.1, timeout is 2 seconds:
Packet sent with a source address of 24.0.0.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2#ping 172.36.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.36.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2#ping 172.36.0.10 source 24.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.36.0.10, timeout is 2 seconds:
Packet sent with a source address of 24.0.0.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R2#ping 172.36.0.10 source 23.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.36.0.10, timeout is 2 seconds:
Packet sent with a source address of 23.0.0.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R2#ping 172.36.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.36.0.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R2#
```